F18CF – Linear Algebra

Tutorial Sheet 5: Linear Maps I

Getting started.

1. Which of the following are linear transformations ?

   (a) $T_1 : \mathbb{R}^2 \to \mathbb{R}^2, \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+y \\ x-y \end{pmatrix}$

   (b) $T_2 : \mathbb{R}^2 \to \mathbb{R}^2, \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+y \\ x \times y \end{pmatrix}$

   (c) $T_3 : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_{2n}(\mathbb{R}), \quad P(x) \mapsto P(x)^2$

   (d) $T_4 : \mathcal{P}(\mathbb{R}) \to \mathbb{R}, \quad P(x) \mapsto \int_0^1 P(t)e^t \mathrm{d}t$

2. For each of the following linear maps, determine the associated matrix in the given basis:

   (a) $T : \mathbb{R}^2 \to \mathbb{R}^2, \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x-y \\ 2x+y \end{pmatrix}$, in the basis $(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix})$ of $\mathbb{R}^2$.

   (b) $T : \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R}), \quad P(x) \mapsto x^3P(2) - xP'(x)$, in the basis $(1, x, x^2, x^3)$ of $\mathcal{P}_3(\mathbb{R})$.

   (c) $T : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R}), \quad P(x) \mapsto P'(x)$ in the basis $(1, x, \ldots, x^n)$ of $\mathcal{P}_n(\mathbb{R})$.

3. For each of the following matrices, determine whether it is invertible, and if so compute its inverse:

$$\text{(a) } A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & -1 & -2 \\ 2 & -9 & 0 \end{pmatrix}, \text{ (b) } B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix}.$$

4. Show that the matrix $A = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}$ is invertible, and compute its inverse.
   Compute the solution of the system of equations $A\vec{x} = \vec{y}$ (in variable $\vec{x}$) for the following values of $\vec{y} \in \mathbb{R}^4$:

$$\text{(a) } \vec{y} = (1,1,1,1)^T \text{ (b) } \vec{y} = (1,2,3,4)^T.$$

Problems.

5. **Polynomial equations.** We consider the following map
$$T : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R}), P(x) \mapsto \frac{P(x) + P(x+1)}{2}.$$
   (a) Show that T is a linear map, and compute its matrix in the basis $\mathcal{B} := (1, x, x^2)$.
   (b) Show that T is invertible and compute $[T]_{\mathcal{B}}^{-1}$.
   (c) Use the previous questions to find the unique polynomials $P(x), Q(x) \in \mathcal{P}_2(\mathbb{R})$ such that
$$\frac{1}{2}(P(x) + P(x+1)) = x^2, \qquad \frac{1}{2}(Q(x) + Q(x+1)) = x^2 + x + 1.$$

6. **Solutions of a homogeneous linear differential equation.** We consider the set W of functions $y \in \mathcal{C}^\infty(\mathbb{R})$ such that
$$y^{(3)} + y^{(2)} + y' + y = 0.$$
   (a) Show that W is a subspace of $\mathcal{C}^\infty(\mathbb{R})$.
   (b) We introduce the map
$$T : W \to \mathbb{R}^3, \quad y \mapsto \big(y(0), y'(0), y^{(2)}(0)\big)^T$$
   that associates to a solution y its vector of initial conditions. Show that T is a linear map.
   (c) Show that the functions $\cos(x),\ \sin(x),\ e^{-x}$ belong to W, and show that their images $T(\cos(x)),\ T(\sin(x)),\ T(e^{-x})$ form a basis of $\mathbb{R}^3$.
   (d) Find a linear combination y of $\cos(x),\ \sin(x),\ e^{-x}$ that is a solution of the differential equation for the following initial condition:
$$y(0) = 3, \quad y'(0) = -2, \quad y^{(2)}(0) = -1.$$

7. **Finding a basis of a span.** The goal of this exercise is to (finally) justify the algorithm for finding a basis of a span of vectors of $\mathbb{R}^n$. Let $\vec{v}_1, \dots, \vec{v}_k$ be a family of vectors of $\mathbb{R}^n$ and let $W := \operatorname{span}(\vec{v}_1, \dots, \vec{v}_k)$. Let B be the $n \times k$ matrix whose columns are $\vec{v}_1, \dots, \vec{v}_k$, and let C be the matrix obtained from B after reducing it to row echelon form.
   (a) Let B' be a matrix obtained by performing some row operation on B. For each of the standard row operations (row exchange, row multiplication, and row addition), find an invertible $n \times n$ matrix E such that $B' = EB$. Deduce that there exists an invertible $n \times n$ matrix A such that $C = AB$.
   (b) Let ℓ be the number of pivots in C, and let V be the subspace of $\mathbb{R}^n$ consisting of vectors whose last $n - \ell$ components are zero. Show that $A\vec{x} \in V$ for every $\vec{x} \in W$.
   (c) Notice that $A\vec{v}_1, \dots, A\vec{v}_k$ are exactly the columns of $C = AB$. Let $\vec{u}_1, \dots, \vec{u}_\ell$ be the sub-family of $\vec{v}_1, \dots, \vec{v}_k$ such that the vectors $A\vec{u}_1, \dots, A\vec{u}_\ell$ contains all the pivots of C. Show that the vectors $A\vec{u}_1, \dots, A\vec{u}_\ell$ form a basis of V.
   (d) Deduce that the vectors $\vec{u}_1, \dots, \vec{u}_\ell$ form a basis of W.